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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Sixth Semester B. Tech (CBCS) Degree Examinations

May 2025

EC3006-1 - VLSI DESIGN

(For EC)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS**20 Marks**

1. **The primary objective of VLSI technology is to:**
A) Decrease circuit complexity
B) Increase circuit speed
C) Increase circuit density
D) Decrease power consumption
2. **Which VLSI design style uses pre-defined logic gates and flip-flops?**
A) Full custom design
B) Semi-custom design
C) Gate array design
D) Standard cell design
3. **The process of converting a hardware description language (HDL) into a gate-level netlist is called:**
A) Behavioral synthesis
B) RTL synthesis
C) Logic synthesis
D) Physical synthesis
4. **Transmission gate _____**
A) Is a bidirectional switch
B) Is a unidirectional switch
C) Consists of a PMOS and a NMOS switch, only one switch is ON at a time
D) Is an inverter
5. **Which of the following is a key factor that affects the speed of VLSI circuits?**
A) Chip packaging
B) Wire length and capacitance
C) Design complexity
D) Transistor color
6. **In VLSI layout, which layer is responsible for connecting the different transistors on a chip?**
A) Active layer
B) Polysilicon layer
C) Metal layer
D) Oxide layer
7. **What does the term "via" refer to in VLSI layout?**
A) A type of transistor used in the design
B) A vertical connection between different layers of metal
C) A method of testing the performance of the chip
D) A type of logic gate used in the design
8. **In constant field MOSFET scaling, what is the key characteristic of the electric field in the channel?**
A) The electric field increases as the gate length decreases
B) The electric field remains constant despite scaling
C) The electric field decreases as the gate length decreases
D) The electric field decreases, but voltage increases
9. **The power dissipation in a CMOS inverter can be minimized by**
A) Increasing the supply voltage
B) Reducing the capacitance at the output
C) Increasing the transistor size
D) Using higher threshold voltage transistors

10. A CMOS inverter has a supply voltage of 1.2V, an output capacitance of 15 pF, and a switching frequency of 50 MHz. What is the dynamic power dissipation?
 A) 0.054 mW B) 0.108 mW
 C) 0.216 mW D) 0.432 mW
11. Path effort is given by _____
 A) $d = h + p$ B) $\hat{D} = NF^{1/N} + P$
 C) $F = GH$ D) $\hat{f} = F^{1/N} = (GH)^{1/N}$
12. Logical effort of NAND3 gate is _____
 A) 5/3 B) 7/3
 C) 4/3 D) 2/3
13. What is the primary benefit of Cascode Voltage Switch Logic (CVSL)?
 A) It eliminates dynamic power consumption completely B) It provides the benefits of ratioed circuits without static power consumption
 C) It achieves the fastest switching among all CMOS logic types D) It reduces transistor count in all logic gates
14. _____ is a type of ratioed logic
 A) Dynamic CMOS B) Tri-state circuit
 C) pseudo nMOS D) Domino Logic
15. Considering pseudo-nMOS, for N inputs, circuit requires _____ FETs
 A) N+2 B) N+1
 C) N(N+1) D) 2N
16. A flip-flop can store _____
 A) One bit of data B) Two bits of data
 C) Any number of bits of data D) Three bits of data
17. ATPG stands for:
 A) Aligned Test Parity Generator B) Automated Test Pattern Generator
 C) Attenuated Transverse wave Pattern Generation D) Automated Testing Phase Generation
18. The ease with which the controller determines signal value at any node by setting input values is known as:
 A) Controllability B) Observability
 C) Testability D) Manufacturability
19. The stuck-at model is a _____ fault model.
 A) Logical fault B) Equivalent fault
 C) Recurring fault D) Simple fault
20. In test mode, storage elements are connected as
 A) Parallel shift register B) Serial shift register
 C) Combiners D) Buffer

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit - I		Marks	BT*	CO*	PO*
1.	a) Sketch a transistor level schematic for a compound level CMOS Logic given by the function $Y = \overline{AB + C} \cdot (A + B)$. Find pFET array and then find nFET array, hence write the complete network.	6	L*3	1	1
	b) Implement 2 input OR gate using transmission gates.	4	L2	1	1
	c) Draw the stick diagram and layout of a 2 input CMOS NAND gate	6	L2	2	1
2.	a) Sketch a transistor level schematic for a compound level CMOS Logic given by the function $Y = (\overline{AB + C}) \cdot \overline{D}$. Find pFET array and then find nFET array, hence write the complete network.	6	L3	1	1
	b) Explain in brief the different types of VLSI design styles.	4	L2	1	1

- c) Briefly discuss constant field scaling and its effect on the current and power characteristics of MOSFET. 6 L2 2 1
3. a) With a neat Figure, explain in detail the VLSI Design Flow. 8 L2 1 1
- b) Draw the transistor level schematic and optimized layout by determining the Euler's path for a CMOS circuit given by the function $F = \overline{A(D + E)} + \overline{BC}$ 8 L3 2 1

Unit – II

4. a) Using the technique of logical effort, determine the optimum sizing of each stage in the path from A to B in the circuit shown in **Figure 4(a)**.

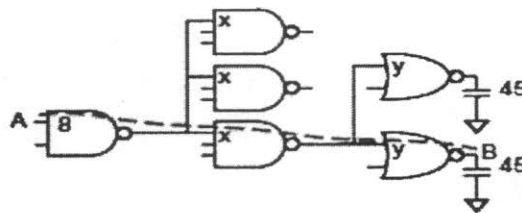


Figure 4(a)

- b) With relevant sketches explain the concept of CVSL. 8 L3 3 1
5. a) With neat diagram, explain the DC transfer characteristics of CMOS inverter. Also show the dependence of slope of VTC on the device ratio. 8 L3 3 1
- b) With relevant sketches explain the concept of Ratioed Logic Circuits. 8 L2 4 1
6. a) Derive the expression for average power dissipation of a CMOS Inverter. 8 L2 3 1
- b) With gate level diagram, explain the working of synchronous resettable CMOS D flip flop. 8 L2 4 1

Unit – III

7. a) Define the following terms with respect to testing. (i) Observability (ii) Controllability (iii) Repeatability (iv) Survivability 8 L2 5 1
- b) Briefly explain Domino CMOS Logic with an example. 8 L2 5 1
8. a) With a neat sketch, explain pre-charge and evaluation modes of dynamic CMOS Logic. Also illustrate the cascading problem in dynamic CMOS Logic. 8 L2 5 1
- b) What is Design for Testability? With a neat block diagram, explain the Built-In-Self-Test approach of Design for Testability. 8 L2 5 1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
